The ABBA_semiimplicit method

Implicit physical step and explicit application of its Jacobian

Adaptation to the GC2D problem with Hairer's symmetric projection

1 Purpose of the method

The ABBA_semiimplicit method advances two objects simultaneously:

$$z_n \in \mathbb{R}^2, \qquad \xi_n \in \mathbb{R}^2,$$

where z_n is the physical state and ξ_n is an infinitesimal perturbation or tangent vector. The state is computed using the complete implicit ABBA method,

$$z_{n+1} = \Psi_{h,t_n}(z_n),$$

whereas the tangent vector is updated by applying the exact Jacobian of that same step:

$$\boxed{\xi_{n+1} = D\Psi_{h,t_n}(z_n)\,\xi_n.}$$

The term *semi-implicit* is used here in a precise sense:

- the trajectory $z_n \mapsto z_{n+1}$ requires solving for the implicit multiplier μ_n;
- once μ_n has converged and the ABBA stages are known, the tangent dynamics is updated by means of a 2×2 linear system.

Essential distinction

The Jacobian formula does not allow the physical step to be replaced by

$$z_{n+1} = D\Psi_{h,t_n}(z_n) z_n.$$

This equality is false for a general nonlinear map. The correct application is

$$\delta z_{n+1} = D\Psi_{h,t_n}(z_n)\,\delta z_n.$$

Therefore, z_{n+1} is first computed with the implicit method, and its Jacobian is then applied to ξ_n.

2 GC2D system and Jacobian of the vector field

The physical state is $z = (X, Y)^T$ and satisfies

$$\dot{z} = f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The spatial Jacobian of the vector field is

$$W(z,t) := D_z f(z,t) = J_0 \nabla_z^2 \phi(z,t) = \begin{pmatrix} -\phi_{XY} & -\phi_{YY} \\ \phi_{XX} & \phi_{XY} \end{pmatrix}_{(z,t)}.$$

Since the Hessian of ϕ is symmetric,

$$W(z,t)^T J_0 + J_0 W(z,t) = 0.$$

3 Implicit ABBA physical step

Let $s = h/2$. For a known state z_n and a provisional value $\mu \in \mathbb{R}^2$, the two copies are initialized as

$$u^{(0)} = z_n + \mu, \qquad v^{(0)} = z_n - \mu.$$

The four A–B–B–A stages are then applied:

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \tag{1a}$$
$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \tag{1b}$$
$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \tag{1c}$$
$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \tag{1d}$$

The multiplier for the step is the root $\mu_n = \mu_h(z_n)$ of the residual

$$\boxed{\mathcal{R}_n(z_n, \mu) := u^{(2)}(z_n, \mu) - v^{(2)}(z_n, \mu) + 2\mu = 0.} \tag{2}$$

Once the implicit iteration has converged, the next physical state is

$$\boxed{z_{n+1} = u^{(2)} + \mu_n = v^{(2)} - \mu_n.} \tag{3}$$

The two expressions in (3) coincide because $\mathcal{R}_n(z_n, \mu_n) = 0$.

4 Construction of the physical Jacobian

All quantities in this section are evaluated *after* the converged root μ_n has been obtained. Define

$$W_1 = W(v^{(0)}, t_n), \qquad W_2 = W(u^{(1)}, t_n),$$
$$W_3 = W(u^{(1)}, t_{n+1}), \qquad W_4 = W(v^{(2)}, t_{n+1}),$$

and

$$S := W_2 + W_3.$$

The Jacobian of the four ABBA stages, regarded as a map on the duplicated phase space, is written in block form as

$$D\Phi^{\mathrm{ABBA}}_{h,t_n} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where

$$A = I_2 + s^2 W_4 S, \qquad B = s(W_1 + W_4) + s^3 W_4 S W_1,$$
$$C = sS, \qquad D = I_2 + s^2 S W_1.$$

To account for the implicit dependence $\mu_n = \mu_h(z_n)$, introduce the four 2×2 matrices

$$\begin{aligned} \mathcal{K} = 4I_2 &- s(W_1 + W_2 + W_3 + W_4) \\ &+ s^2(W_4 S + S W_1) - s^3 W_4 S W_1, \end{aligned} \tag{4}$$

$$\begin{aligned} \mathcal{L} = s&(W_1 - W_2 - W_3 + W_4) \\ &+ s^2(W_4 S - S W_1) \\ &+ s^3 W_4 S W_1, \end{aligned} \tag{5}$$

$$
\begin{aligned}
\mathcal{P} &= I_2 + s(W_1 + W_4) \\
&\quad + s^2 W_4 S + s^3 W_4 S W_1,
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
\mathcal{Q} &= 2I_2 - s(W_1 + W_4) \\
&\quad + s^2 W_4 S - s^3 W_4 S W_1.
\end{aligned} \tag{7}
$$

Here,

$$\mathcal{K} = D_\mu \mathcal{R}_n, \qquad \mathcal{L} = D_z \mathcal{R}_n.$$

Differentiating the identity

$$\mathcal{R}_n(z, \mu_h(z)) = 0$$

with respect to z gives

$$\mathcal{L} + \mathcal{K} D\mu_h(z) = 0,$$

and, provided that $\mathcal{K}$ is invertible,

$$D\mu_h(z_n) = -\mathcal{K}^{-1}\mathcal{L}.$$

Consequently, the exact Jacobian of the implicit physical step is

$$\boxed{J_n := D\Psi_{h,t_n}(z_n) = \mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}.} \tag{8}$$

The matrix $\mathcal{K}$ in (4) is also the Jacobian of the residual used in Newton's method. However, $\mathcal{K}$ is not the physical Jacobian: the latter is J_n and also contains $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{L}$.

5 Applying the Jacobian without computing an inverse

In an implementation, $\mathcal{K}^{-1}$ should not be formed explicitly. To construct the entire Jacobian, solve the system with two right-hand sides:

$$\boxed{\mathcal{K}\mathcal{T} = \mathcal{L}, \qquad J_n = \mathcal{P} - \mathcal{Q}\mathcal{T}.} \tag{9}$$

Since $\mathcal{K} \in \mathbb{R}^{2\times 2}$, its factorization is inexpensive and can be reused for the two columns of $\mathcal{L}$.

If only a tangent vector ξ_n is to be propagated, it is not even necessary to form $\mathcal{T}$ or J_n. From (8),

$$J_n \xi_n = \mathcal{P}\xi_n - \mathcal{Q}\mathcal{K}^{-1}(\mathcal{L}\xi_n).$$

Therefore, compute successively

$$b_n = \mathcal{L}\xi_n, \qquad \mathcal{K} y_n = b_n,$$

and finally

$$\boxed{\xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q} y_n.} \tag{10}$$

The only implicit part of (10) is a 2×2 linear system; no new nonlinear equation appears.

5.1 Explicit formula for the 2×2 system

Let

$$\mathcal{K} = \begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}, \qquad b_n = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \qquad \Delta_{\mathcal{K}} = k_{11}k_{22} - k_{12}k_{21}.$$

If $\Delta_{\mathcal{K}} \neq 0$, the solution of $\mathcal{K} y_n = b_n$ is

$$\boxed{y_n = \frac{1}{\Delta_{\mathcal{K}}} \begin{pmatrix} k_{22}b_1 - k_{12}b_2 \\ -k_{21}b_1 + k_{11}b_2 \end{pmatrix}.} \tag{11}$$

This expression makes it possible to apply (10) without a general-purpose matrix solver.

6 One-step algorithm

Given t_n, z_n, ξ_n, and h, one step of `ABBA_semiimplicit` is as follows:

1. Solve the nonlinear equation (2) to obtain μ_n. Each residual evaluation executes the four ABBA stages in (1).
2. With the converged root, recompute or retain the stages $u^{(0)}, v^{(0)}, u^{(1)}, v^{(1)}, v^{(2)}, u^{(2)}$.
3. Compute the new physical state using (3).
4. Evaluate W_1, W_2, W_3, W_4 at the four stage points and form $S = W_2 + W_3$.
5. Construct $\mathcal{K}$, $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{Q}$ using (4)–(7).
6. Compute $b_n = \mathcal{L}\xi_n$ and solve $\mathcal{K}y_n = b_n$.
7. Apply the Jacobian through $\xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n$.
8. Advance the time: $t_{n+1} = t_n + h$.

The output of the step is the pair

$$\boxed{(z_{n+1}, \xi_{n+1}).}$$

7 Propagation of the accumulated Jacobian

To obtain the derivative of the numerical solution with respect to the complete initial condition, propagate a matrix $M_n \in \mathbb{R}^{2\times 2}$ instead of a single vector. Initialize

$$M_0 = I_2$$

and, after each physical step, update it according to

$$\boxed{M_{n+1} = J_n M_n.} \tag{12}$$

Thus,

$$M_N = J_{N-1} J_{N-2} \cdots J_1 J_0 = \frac{\partial z_N}{\partial z_0}.$$

The order of the products is important: the Jacobian of the most recent step always multiplies from the left.

The matrix update can also be performed without forming J_n. Solve

$$\mathcal{K}Y_n = \mathcal{L}M_n$$

for the two right-hand sides and set

$$\boxed{M_{n+1} = \mathcal{P}M_n - \mathcal{Q}Y_n.} \tag{13}$$

8 Symplecticity of the tangent propagation

If the projected ABBA step is solved exactly, its physical Jacobian satisfies

$$J_n^T J_0 J_n = J_0.$$

In two dimensions, this identity is equivalent to

$$\det J_n = 1.$$

Therefore, the application $\xi_{n+1} = J_n \xi_n$ is a symplectic linear transformation at each step. In addition, the accumulated product (12) is also symplectic:

$$M_N^T J_0 M_N = J_0, \qquad \det M_N = 1.$$

In numerical computations, it is useful to monitor

$$\varepsilon_{\mathrm{sp},n} := \left\| J_n^T J_0 J_n - J_0 \right\|, \qquad \varepsilon_{\det,n} := |\det J_n - 1|.$$

These quantities will not be exactly zero because of roundoff, the tolerance used to solve (2), and errors in the derivatives of ϕ.

9 Interpretation in terms of two nearby trajectories

Let $z_n^\varepsilon = z_n + \varepsilon \xi_n$, with ε small. Differentiability of the implicit map gives

$$\Psi_{h,t_n}(z_n^\varepsilon) = \Psi_{h,t_n}(z_n) + \varepsilon D\Psi_{h,t_n}(z_n)\xi_n + \mathcal{O}(\varepsilon^2).$$

Since $z_{n+1} = \Psi_{h,t_n}(z_n)$ and $\xi_{n+1} = D\Psi_{h,t_n}(z_n)\xi_n$, it follows that

$$z_{n+1}^\varepsilon = z_{n+1} + \varepsilon \xi_{n+1} + \mathcal{O}(\varepsilon^2).$$

Thus, `ABBA_semiimplicit` simultaneously computes the reference trajectory and the linear evolution of an infinitesimal separation from it.

10 Implementation-ready summary

At each time step, perform the two updates

$$\boxed{\begin{array}{ll} z_{n+1} = \Psi_{h,t_n}(z_n) & \text{using ABBA and the implicit root } \mu_n, \\ \xi_{n+1} = \left(\mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L}\right)\xi_n & \text{using a } 2 \times 2 \text{ linear system.} \end{array}}$$

The recommended form for the second line is

$$b_n = \mathcal{L}\xi_n, \qquad \mathcal{K} y_n = b_n, \qquad \xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q} y_n.$$

The physical state is not obtained by multiplying z_n by the Jacobian. The Jacobian is applied to the tangent vector or, equivalently, to the accumulated Jacobian with respect to the initial condition.